

FERRY CAPACITY UTILIZATION & OPERATIONAL EFFICIENCY ANALYTICS SYSTEM

1. INTRODUCTION

Public ferry transportation plays an important role in providing safe and efficient travel between Toronto's mainland and the Toronto Islands. During peak seasons, ferry terminals experience high passenger demand, while off-season periods often result in underutilized ferry capacity. This project analyzes historical ferry ticket sales and redemption data to evaluate operational efficiency, identify congestion periods, and support data-driven decision-making for ferry management.

2. PROBLEM STATEMENT

FERRY OPERATIONS CURRENTLY LACK A STRUCTURED ANALYTICAL SYSTEM TO MONITOR CAPACITY UTILIZATION, DETECT CONGESTION, AND IDENTIFY IDLE OPERATING PERIODS. THE OBJECTIVE OF THIS PROJECT IS TO ANALYZE FERRY TICKET DATA AND PROVIDE INSIGHTS THAT HELP OPTIMIZE FERRY SCHEDULING, IMPROVE PASSENGER EXPERIENCE, AND REDUCE OPERATIONAL COSTS.

3. OBJECTIVE

- Primary Objectives
 - Measure ferry capacity utilization.
 - Identify peak and idle operating periods.
 - Evaluate operational efficiency using historical ticket data.
- Secondary Objectives
 - Support resource allocation decisions.
 - Improve passenger comfort and safety.
 - Assist seasonal planning for ferry operations.

4. DATASET DESCRIPTION

[9:32 pm, 30/07/2026] DISHIKA: The dataset contains historical ferry ticket records from 2015–2025 with the following attributes:

Column	Description
_id	Unique record identifier
Timestamp	15-minute interval timestamp
Sales Count	Number of tickets sold
Redemption Count	Number of tickets redeemed

5. DATA PREPROCESSING

The following preprocessing steps were performed:

- Loaded the dataset using *Pandas*.
- Converted timestamps to *datetime* format.
- Checked for missing values and duplicate records.
- Created new features such as:

Year

Month

Weekday

Season

Total Activity Load

Redemption Pressure Ratio

Operational Load Index

Idle Capacity Indicator

6.EXPLORATORY DATA ANALYSIS (EDA)

EDA included:

- *Sales Count distribution*
- *Redemption Count distribution*
- *Sales vs Redemption comparison*
- *Monthly activity analysis*
- *Hourly activity trends*
- *Weekday vs Weekend comparison*
- *Seasonal activity analysis*
- *Operational Load Index distribution*
- *Idle Capacity analysis*
- *Correlation analysis*
- *Boxplots for outlier detection*
- *busiest hours*

7.KPI ANALYSIS

The following KPIs were calculated:

- *Capacity Utilization Ratio*
- *Congestion Pressure Index*
- *Idle Capacity Percentage*
- *Peak Strain Duration*
- *Operational Variability Score*

These KPIs provide insights into ferry efficiency and operational performance

8.TIME SERIES ANALYSIS

Time-series analysis was performed using daily, hourly, and monthly aggregations. Trends were visualized to identify seasonal variations, peak operating periods, and long-term changes in ferry utilization.

9.DASHBOARD DEVELOPMENT

An interactive Streamlit dashboard was developed featuring:

- 1.Dataset preview*
- 2.KPI summary cards*
- 3.Hourly activity visualization*
- 4.Monthly activity trends*
- 5.Capacity utilization metrics*

The dashboard enables quick exploration of ferry operational performance.

10.RESULTS

The analysis showed that:

- Ferry demand varies significantly by season.*
- Peak congestion occurs during high-demand periods.*
- Some operating intervals exhibit low utilization.*
- Capacity utilization patterns can support better scheduling decisions.*

11.RECOMMENDATIONS

- Increase ferry frequency during peak periods.*
- Reduce unnecessary ferry operations during low-demand intervals.*
- Continuously monitor operational KPIs.*
- Use data-driven scheduling to improve efficiency and reduce costs*

12.CONCLUSION

This project successfully analyzed ferry ticket activity to evaluate operational efficiency. By combining data preprocessing, exploratory analysis, KPI calculations, time-series analysis, and an interactive dashboard, the project provides valuable insights for improving ferry scheduling and resource allocation. The proposed analytical framework can assist

transportation authorities in making informed operational decisions and enhancing passenger service.